

Project Initialization and Planning Phase

Date	6 july , 2026
Team ID	SWTID-2026-8420
Project Title	Credit Card Approval Prediction using Machine Learning
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) Report

This project automates the credit card approval process using machine learning. It analyses applicant financial and demographic information to predict whether a credit card application should be approved or rejected. The solution reduces manual effort, improves consistency, speeds up decision-making, and minimizes the risk of human error.

Project Overview	
Objective	To develop a machine learning system that predicts credit card approval chances quickly and accurately based on customer financial details.
Scope	The project focuses on automating the credit card approval process using ML models by analyzing factors like income, credit score, age, and debt.
Problem Statement	
Description	Traditional credit card approval processes are slow and require manual verification, which may cause delays and inaccurate decisions.
Impact	The solution improves approval speed, reduces manual effort, minimizes errors, and provides better customer experience.
Proposed Solution	
Approach	Use machine learning algorithms to analyze applicant details and predict whether the credit card application will be approved or rejected.
Key Features	- Fast approval prediction - Automated decision-making - Accurate risk assessment - User-friendly prediction interface - Real-time results

Resource Requirements

Resource Type	Description	Specification/ Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel Core i5/i7 Processor (4+ Cores)
Memory	RAM specifications	8 GB RAM (Minimum), 16 GB Recommended
Storage	Disk space for datasets, trained models, and application files	512 GB SSD / 1 TB HDD
Software		
Frameworks	Python Web Framework	Flask
Libraries	Machine Learning and Data Processing Libraries	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, XGBoost
Development Environment	IDE	Jupyter Notebook, Visual Studio Code, PyCharm
Data		
Data	Source, Size, Format	Credit Card Approval Dataset (Kaggle), CSV Format, ~20,000+ Applicant Records